INCH-POUND

A-A-59101B
14 July 2004
SUPERSEDING
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# COMMERCIAL ITEM DESCRIPTION
# WATTMETER
# METER, AUDIO POWER 20 Hz to 20 KHz

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID), describes Wattmeter (Audio Power Meter). This CID is meant as a minimum requirement for the audio power meter in which only those manufacturers that meet or surpass the following requirements are to be supplied per this CID. This wattmeter is used to measure the output power of audio devices in the frequency range of 20 Hz to 20KHz.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

2.1 Classification. Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet this description. The equipment shall be Class 3 in accordance with MIL-PRF-28800, except as specified herein.

2.2 Safety and Environmental. The audio power meter shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein except as specified herein.

2.2.1 Temperature Operating. The audio power meter shall conform to its specified performance and accuracy while being operated in a temperature range of 0°C to +50°C.

Comments, suggestions, or questions on this document should be addressed to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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2.2.3 Temperature Non-operating. The audio power meter shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of -10°C to +60°C.

2.3 Electrical Power Sources. The equipment shall operate from nominal line voltages of 110 volts AC (±10%) and frequencies of 47 Hz to 63 Hz. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein.

2.4 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 2000 hours of operation with a statistical certainty of 95%. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion. Records of the actual equipment model shall be used, unless the Government authorizes the substitution of records from a similar model, due to the lack of sufficient supporting documentation on the offered equipment. The manufacturer shall be able to reasonably establish the claim that the offered equipment shall provide reliability similar to the model used to provide substitute records.

2.5 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours. No special tools shall be required for alignment or adjustment except those furnished with the Transmission Line Test Set.

2.6 Performance Requirements. The equipment shall meet all requirements specified herein, under all combinations of input power conditions, output signal conditions and operating service condition as specified in MIL-PRF-28800.

2.6.1 Frequency Range. The minimum frequency range of the audio power meter shall be 20 Hz to 20 kHz.

2.6.2 Power Range. The power meter shall have as a minimum, a range from 300 microwatts to 20 Watts.

2.6.3 Power Accuracy. The minimum power accuracy of the audio power meter shall be ±7.5% Full Scale Deflection (FSD).

2.6.4 Impedance. The impedance range of the audio power meter shall include a minimum range of 10 ohms to 600 ohms.

2.6.5 Impedance Accuracy. The impedance accuracy of the audio power meter shall be a minimum of ±7% for 70 Hz to 5 kHz.

2.6.6 Interface. The audio power meter shall include front panel controls.



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2.6.7 Connectors. The audio power meter shall have three binding posts on the front panel. Two of the binding posts shall be for inputs, and one binding post shall be for chassis ground.

2.6.8 Front Panel Display. The audio power meter shall have a meter/display on the front panel that displays power reading in linear mode: milliwatts/Watts (dependent on range selected) and in logarithmic mode: dBm.

2.6.9 Power Range Switch. The audio power meter shall have a switch on the front panel for the selection of an appropriate power range for unit under test power output measurement.

2.6.10 Impedance Selection Switch. The audio power meter shall have a switch on the front panel to select the impedance (load) that will be applied to the unit under test.

2.6.11 Physical Characteristics. The equipment's dimensions and weight shall be consistent with current commercial offerings. The equipment shall have the following physical characteristics.

2.6.11.1 Dimensions. The width and depth of the equipment shall be compatible with mounting in an ANSI/EIA-310-D standard equipment rack. The height shall be compatible with two standard height units, 2U. One unit or "U" is defined by ANSI/EIA-310-D to be 1.75 inches or 44.45 mm.

2.6.11.2 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed 20 pounds (or 9.1 kilograms).

If the depth of the equipment is greater than 24 inches the maximum allowable weight shall be reduced by 33 percent.

2.6.12 Manuals. The equipment shall be delivered with operator, maintenance, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCE CONFORMANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.



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4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.3 ANSI/EIA standards can be obtained from the Electronics Industry Association, Engineering Department, 2001 Pennsylvania Ave., N.W., Washington, D.C., 20006. Phone: 1-800-854-7179 (USA and Canada).

MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
Air Force - 84

Agent:
Air Force - 99

(Project 6625-F089)

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